Solution: Let, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, where $x_1, x_2 \in R$.

$\therefore \begin{bmatrix} x_1\\x_2\end{bmatrix} = \begin{bmatrix} x_1+0\\0+x_2\end{bmatrix} = \begin{bmatrix}x_1\\0\end{bmatrix}+\begin{bmatrix}0\\x_2\end{bmatrix}$ (Vector addition)

$= \begin{bmatrix}x_1\cdot 1\\x_1\cdot 0\end{bmatrix}+\begin{bmatrix}x_2\cdot 0\\x_2\cdot 1\end{bmatrix} = x_1\begin{bmatrix}1\\0\end{bmatrix}+x_2\begin{bmatrix}0\\1\end{bmatrix}$ (Scalar multiplication)

$= x_1e_1+x_2e_2$ $\quad \therefore T(x)=T(x_1e_1+x_2e_2)=T(x_1e_1)+T(x_2e_2)$ (Condition 1)

$= \boxed{x_1T(e_1)+x_2T(e_2) \text{ (Condition 2)}} = x_1\begin{bmatrix}5\\-7\\2\end{bmatrix}+x_2\begin{bmatrix}-3\\8\\0\end{bmatrix} = \begin{bmatrix}5x_1-3x_2\\-7x_1+8x_2\\2x_1+0\end{bmatrix}$ □

$$\boxed{T(x) = \begin{bmatrix}T(e_1) & T(e_2)\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix} = Ax}$$ (By defn. of matrix eqn. $Ax=b$)

Theorem 10: Let $T: R^n \to R^m$ be a linear transformation. Then there exists a unique matrix A such that $T(x)=Ax$ for all x in R^n. In fact, A is the $m\times n$ matrix whose jth column is the vector $T(e_j)$, where e_j is the jth column of the identity matrix in R^n:

$$A = \begin{bmatrix}T(e_1) & \cdots & T(e_n)\end{bmatrix} \quad (3)$$

→ Why? Prove it

Proof: • We can write $x = I_nx = \begin{bmatrix}e_1 & e_2 & \cdots & e_n\end{bmatrix}x = x_1e_1+x_2e_2+\cdots+x_ne_n$.

$T(x)=T(x_1e_1+x_2e_2+\cdots+x_ne_n)=T(x_1e_1)+T(x_2e_2)+\cdots+T(x_ne_n)$ (Condition 1)

$=x_1T(e_1)+x_2T(e_2)+\cdots+x_nT(e_n)$ (Condition 2)

$= \begin{bmatrix}T(e_1) & T(e_2) & \cdots & T(e_n)\end{bmatrix}\begin{bmatrix}x_1\\x_2\\ \vdots\\x_n\end{bmatrix}$ (By defn. of matrix equation $Ax=b$)

$\therefore A = \begin{bmatrix}T(e_1) & T(e_2) & \cdots & T(e_n)\end{bmatrix}$ □

We must prove the uniqueness of A in Exercise 33

• The matrix A in (3) is called the standard matrix for the linear transformation T.